Frozen UniSpeech-SAT and WavLM for Speech Emotion Recognition: Unseen Speakers, Noise and Noise-Augmented Classifier Training

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Abstract

Speech emotion recognition (SER) faces considerable challenges when processing speech from unfamiliar speakers or speech affected by background noise. This study asks whether UniSpeech-SAT Base+ and WavLM Base+, two self-supervised models with different pretraining objectives, differ in robustness to unseen speakers and acoustic noise when used as frozen feature extractors. Mean-pooled 768-dimensional embeddings from each encoder were classified by identical multilayer perceptrons on 672 RAVDESS recordings (neutral, happy, sad, angry), with an actor-independent split reserving 112 recordings from four unseen speakers for testing. Macro-F1 was the primary metric, supported by accuracy and confusion matrices to expose class-level errors.

On clean speech, WavLM scored higher (macro-F1 0.8186 vs 0.7573). However, when DEMAND kitchen, cafeteria and traffic noise were added at 10 dB and 0 dB signal-to-noise ratio, both clean-trained classifiers degraded sharply, and neither model was consistently more robust; for example, UniSpeech-SAT outperformed WavLM in kitchen noise. Under cafeteria noise, both classifiers labelled every recording “sad”, giving a macro-F1 of 0.1111 while accuracy (28.6%) concealed the failure. Training new classifiers on clean and noise-augmented embeddings, with the encoders still frozen, improved macro-F1 in all six noisy conditions, by up to +0.62 for WavLM and +0.56 for UniSpeech-SAT, although the ranking of the two models still varied with noise type and SNR.

These results suggest that clean-speech performance does not predict noisy performance, and that downstream classifier training strongly affects robustness. Conclusions are limited by a single speaker split and by using the same noise recordings for training and testing, so the gains reflect the tested noise sources rather than unseen environments.

1) Introduction:

Speech emotion recognition (SER) seeks to identify emotional expression from acoustic properties such as fundamental frequency, intensity, voice quality and temporal structure. These properties are not determined by emotion alone: they also reflect speaker characteristics, speaking style and recording conditions, and environmental noise can further obscure the cues used for classification. A central challenge is therefore developing representations that remain informative when the speaker or acoustic environment differs from training.

Early SER systems relied on handcrafted features such as MFCCs (Mel Frequency Cepstral Coefficient), pitch and energy, combined with conventional classifiers and, more recently, deep and convolutional networks [8], [9]. Their success depends on whether the chosen features capture emotion-relevant information and remain reliable under changing conditions. Self-supervised learning (SSL) offers an alternative: models pretrained on large amounts of unlabelled speech yield representations that transfer to emotion classification with lightweight classifiers [5], [6]. Atmaja and Sasou [5] found WavLM and UniSpeech-SAT among the strongest SSL representations across five SER datasets.

These two models are pretrained differently. UniSpeech-SAT adds speaker-aware contrastive learning and utterance mixing to strengthen speaker information [1], whereas WavLM combines masked prediction with denoising [2]. It is not obvious how these choices affect SER: speaker-aware features may not generalise to unfamiliar voices, and a denoising objective does not explicitly preserve emotion-specific cues. Existing work leaves this open. Comparisons such as [5] concentrate on clean-speech performance, while noise-robust SER research [7], [12] has focused on augmentation or environment-aware adaptation rather than comparing frozen encoders under identical conditions. To the best of my knowledge, the Base+ variants of UniSpeech-SAT and WavLM have not been directly compared under controlled noise at fixed signal-to-noise ratios (SNRs) on unseen speakers.

This study addresses that gap through three connected experiments, using frozen encoders with mean-pooled embeddings and identical multilayer perceptron (MLP) classifiers: (i) a clean-speech comparison on four unseen RAVDESS actors [3]; (ii) evaluation under kitchen, cafeteria and traffic noise from DEMAND [4] at 10 dB and 0 dB SNR; and (iii) retraining the classifiers on clean and noise-augmented embeddings without modifying the encoders. Performance is assessed using macro-F1, accuracy and confusion matrices.

The scope is deliberately narrow. The study uses one actor-independent split of acted speech, four emotion classes, final-layer embeddings, and noise recordings shared between augmentation and testing. It therefore characterises the behaviour of two representations under the tested conditions, rather than isolating the effect of their pretraining objectives or proving generalisation to unseen environments. The research questions are:

RQ1: How do frozen UniSpeech-SAT Base+ and WavLM Base+ representations compare in clean and noisy speech emotion recognition when evaluated on previously unseen speakers?

RQ2: Can noise-augmented downstream classifier training reduce noise-related degradation without modifying the pretrained encoders?

The remainder of this report reviews related work, describes the methodology, presents results for the three experiments, and discusses their interpretation and limitations.

2) Literature Review

2.1) From handcrafted features to self-supervised representations

Speech emotion recognition (SER) has traditionally relied on handcrafted descriptors such as MFCCs, pitch, energy and zero-crossing rate (ZCR), passed to a trained classifier. Hama Saeed [8] and Alluhaidan et al. [9] show that deep and convolutional classifiers can improve recognition using such features. The limitation is that classifier complexity cannot recover information discarded at the feature stage: MFCCs summarise the short-term spectral envelope but do not explicitly encode longer prosodic patterns, and they are sensitive to speaker and channel effects.

Self-supervised learning (SSL) addresses this by learning representations from large amounts of unlabelled speech, so scarce emotion labels are needed only downstream. Wav2vec 2.0 [14] established the paradigm by masking latent speech frames and training the model to identify the true quantised target among distractors. Pepino et al. [6] showed that wav2vec 2.0 embeddings feeding shallow classifiers perform competitively on IEMOCAP and

RAVDESS, and that combining encoder layers matters. Atmaja and Sasou [5] compared many SSL representations across five SER datasets using a common classifier, with WavLM and UniSpeech-SAT among the strongest. This motivates the choice of these two models, but [5] does not examine how the Base+ variants respond to controlled environmental noise.

2.2) Pretraining objectives: UniSpeech-SAT and WavLM

UniSpeech-SAT and WavLM use different methods to learn speech features during pretraining. UniSpeech-SAT [1] focuses particularly on recognising information about individual speakers. One technique it uses is *utterance mixing*, where speech from another person is added to a recording, encouraging the model to distinguish the original speaker. Although this helps the model learn speaker-related features, it may also cause problems for speech emotion recognition. For example, a classifier might learn patterns associated with particular speakers rather than emotions themselves, making it less accurate when analysing unfamiliar voices. This is why the present study separates speakers between the training and testing datasets.

WavLM [2] takes a different approach by learning to recognise speech even when background noise is present. During pretraining, noise or overlapping speech is added to recordings, and the model learns to predict information from the original clean speech. This technique, known as *masked denoising prediction*, helps WavLM develop features that are less affected by acoustic interference. It also uses a mechanism called *gated relative position bias* to help the model understand how different parts of a speech recording relate to one another over time. However, recognising speech clearly does not necessarily mean recognising emotions accurately, since background noise may still affect important emotional cues. Furthermore, because WavLM and UniSpeech-SAT differ in several aspects of their training and design, differences in their performance cannot be attributed to their pretraining methods alone.

2.3) Robustness of SSL representations and of SER under noise

Domain shift has been investigated in SSL-based automatic speech recognition by Hsu et al. [15], who demonstrated that incorporating target-domain data during pretraining can improve performance under mismatched conditions. However, these findings concern speech recognition rather than emotion recognition, and therefore cannot directly establish whether emotionally relevant SSL representations remain robust to environmental noise. SUPERB [13] supports the frozen-encoder evaluation approach adopted here and includes an emotion recognition task, but it is built around standard benchmark conditions and does not itself test how particular models behave under controlled noise at 10 dB and 0 dB SNR.

Within SER, Barhoumi and BenAyed [7] show that noise addition and related augmentations improve deep classifiers built on conventional acoustic features, but this does not establish the same effect for frozen SSL encoders. Leem et al. [12] improve noise robustness through text-guided environment-aware training, fusing a text description of the acoustic environment with a transformer-based SER model and predicting continuous arousal, valence and dominance using real-world noise. Their method adds information and model adaptation, whereas the present study asks whether changing only a lightweight downstream classifier's training data can recover performance. Because their targets are continuous and ours are four discrete classes, their results are a methodological reference point rather than a numerical benchmark.

2.4) Representation extraction and classifier design

Fine-tuning a full encoder can improve accuracy but confounds the effect of pretraining with the effect of task adaptation. Frozen encoders with an identical lightweight classifier allow a cleaner comparison [6]. Two design choices still affect results. First, Zhu and Sato [11] found that emotion-relevant information is not concentrated in the final layer of models including WavLM, so a final-layer embedding may be suboptimal. Second, temporal mean pooling gives a fixed-length vector from variable-length input but discards temporal order and local prosodic events; attention-based pooling retains more at greater complexity. This study adopts final-layer embeddings, mean pooling and a single-hidden-layer MLP, trading optimal accuracy for a controlled comparison. Specialised emotion encoders such as emotion2vec [10] may outperform general-purpose models, but they would answer a different question from the relative robustness of two general-purpose encoders.

2.5) Evaluation design: RAVDESS, DEMAND and metrics

RAVDESS [3] offers acted emotional speech from 24 actors, which allows a clean actor-independent split but limits conclusions about spontaneous speech. DEMAND [4] provides real multichannel environmental recordings that can be mixed with speech at controlled SNRs; synthetic mixing is reproducible but omits reverberation and microphone effects. Two methodological points follow. Mixing the same noise recordings into both augmented training data and noisy test data shows robustness to those sources, not to unseen environments and because noise can collapse a classifier onto one class, accuracy alone can be misleading when predictions collapse onto a single class, particularly when class frequencies are unequal on imbalanced data, so macro-F1 and confusion matrices are required alongside it.

2.6) Research Gap

Previous studies have established the effectiveness of SSL representations for SER [5], [6] and investigated methods for improving recognition under environmental noise [7], [12]. Leem et al. [12] improved noise robustness through environment-aware adaptation using additional textual descriptions of the acoustic environment, rather than comparing frozen pretrained encoders under identical downstream conditions. Atmaja and Sasou [5] compared many SSL representations across several datasets but, in the sources reviewed, did not examine how the Base+ variants of UniSpeech-SAT and WavLM respond to controlled noise at fixed SNRs.

This study addresses that narrower gap through three connected experiments: (i) a clean unseen-speaker comparison of frozen UniSpeech-SAT Base+ and WavLM Base+ using identical classifiers and actor-independent RAVDESS partitions; (ii) evaluation under matched DEMAND noise at 10 dB and 0 dB SNR; and (iii) noise-augmented classifier training with the encoders unchanged. The contribution is a controlled evaluation and a practical modification motivated by the observed failures, not a new pretraining method. Because the design uses one speaker split and shared noise recordings, the results indicate how these two representations behave under the tested conditions, rather than isolating the effect of their pretraining objectives or demonstrating generalisation to unseen environments.

3) Method:

3.1 Overview and design rationale

Prior SER work shows that lightweight classifiers on pretrained SSL embeddings are effective [5], [6]. Fine-tuning the encoder adds a further variable that complicates comparison between models. Both models were therefore used as frozen feature extractors with identical downstream classifiers, so differences can be examined without introducing encoder fine-tuning as an additional variable. Three experiments were run: (1) clean unseen-speaker evaluation, (2) evaluation of the clean-trained classifiers under DEMAND noise, and (3) retraining the classifiers on clean and noise-augmented embeddings. The implementation used Python, PyTorch, Hugging Face Transformers, Librosa and Scikit-learn.

3.2 Data and splits

The speech portion of RAVDESS [3] (24 actors) was used, retaining four categories: neutral, happy, sad and angry. This gives 672 recordings: 28 per actor, since neutral is recorded only at normal intensity (4 recordings), while the other three emotions have two intensities (8 each). Audio was converted to mono and resampled from 48 kHz to 16 kHz, as required by both encoders. Actors were split by identity: actors 1–16 for training (448 recordings), 17–20 for validation (112) and 21–24 for testing (112). Because RAVDESS numbers its actors so that odd numbers are male and even numbers female [3], each split contains equal numbers of male and female speakers. The test set holds 16 neutral recordings and 32 of each other class. This imbalance motivated class-weighted loss and macro-F1.

3.3 Representation extraction

UniSpeech-SAT Base+ [1] and WavLM Base+ [2] were loaded through Hugging Face Transformers using the pretrained checkpoints microsoft/unispeech-sat-base-plus and microsoft/wavlm-base-plus, respectively, with all model parameters frozen. The final hidden layer's 768-dimensional frame sequence was averaged over time to produce one embedding per recording. Final-layer mean pooling is a deliberate simplification. Zhu and Sato [11] report that intermediate layers can carry more emotion information, and mean pooling discards temporal order. The results therefore describe these representations under this specified downstream design, not their best achievable performance.

3.4 Classifier

Each model used an identical MLP: 768 inputs, one hidden layer of 256 units (ReLU, dropout 0.3) and four outputs. Embedding dimensions were standardised using training-set statistics only. Training used Adam (learning rate 0.001, batch size 32, up to 50 epochs), class-weighted cross-entropy and a fixed seed of 42. The same settings were applied to both models. The checkpoint with the highest validation macro-F1 was retained, so the test set played no role in model selection.

3.5 Noise conditions (Experiment 2)

Real recorded noise from DEMAND [4] was used. Three environments were selected to represent different types of acoustic interference: kitchen (DKITCHEN), cafeteria (PCAFETER) and traffic (STRAFFIC). These provided household noise containing transient events, speech-like background babble and road noise, respectively. The 16 kHz mono recordings from channel ch01.wav were used for each environment.

Each environment was mixed with the speech at 10 dB (moderate interference) and 0 dB (equal speech and noise energy) signal-to-noise ratio (SNR). Noise was scaled using root mean square (RMS) amplitudes according to:

Equation 1. Calculation of the noise scaling factor (α) using the RMS amplitudes of

$$\alpha = \frac{RMS_{speech}}{RMS_{noise} \times 10^{\frac{SNR}{20}}}$$

The noisy waveform was then generated as:

$$Mixture = Speech + \alpha \times Noise$$

This gives the relationship:

$$SNR = 20 \log_{10}\left(\frac{RMS_{speech}}{\alpha \times RMS_{noise}}\right)$$

Equation 3. Calculation of the signal-to-noise ratio (SNR) of the resulting mixture using the RMS amplitudes of speech and noise.

Understanding that SNR represents a ratio rather than an absolute signal level was necessary for implementing the noise-mixing procedure correctly.
For each test recording and environment, a duration-matched noise segment was selected using a randomly generated starting offset. NumPy's random number generator with a fixed seed of 42 ensured that the offsets were reproducible. The same selected segments and starting offsets were used at both SNR levels, and the resulting noisy waveforms were applied identically to both pretrained models. This ensured that differences in classification performance were not caused by differences in the noise inputs. The original clean-trained classifiers were evaluated without modification.

3.6 Noise-augmented training
Each training and validation recording was supplemented with two noisy copies, at 10 dB and 0 dB SNR, using randomly selected DEMAND noise under a reproducible selection procedure. This increased the training set to 1,344 embeddings and the validation set to 336. The architecture and training settings were unchanged, and the checkpoint was selected by mean validation macro-F1 across clean, 10 dB and 0 dB conditions.
Two weaknesses of this design are acknowledged. First, the same three DEMAND recordings were used for augmentation and for testing, so any gain reflects robustness to these noise sources rather than to unseen environments. Second, the training set triples in size, so gains cannot be attributed to noise exposure alone. These are discussed in Section 5 and addressed in the proposed future work.

3.7 Evaluation
Macro-F1 was the primary metric because it weights all four classes equally and exposes class collapse that accuracy can hide. Accuracy, per-emotion F1 and confusion matrices were reported alongside it. All evaluations used the same 112 test recordings. Each configuration was trained once with seed 42, and no confidence intervals were computed, so small score differences should be interpreted cautiously. Embedding-extraction times were recorded in local runs and are implementation-specific, not a controlled hardware benchmark.

4) Results:

All reported results were obtained from the same 112 held-out RAVDESS recordings (actors 21–24). Macro-averaged F1 was used as the primary performance measure, supported by classification accuracy, per-emotion F1 scores and confusion matrices. *Table 1* compares the original classifiers trained on clean embeddings with the new classifiers trained using both clean and noise-augmented embeddings.

Audio condition	UniSpeech-SAT Clean-trained	UniSpeech-SAT Noise-augmented	WavLM Clean-trained	WavLM Noise-augmented
Clean	0.7573	0.7992	0.8186	0.8503
Kitchen (10 dB)	0.3686	0.7516	0.2447	0.8033
Kitchen (0 dB)	0.3813	0.7721	0.1278	0.7055
Cafeteria (10 dB)	0.1111	0.6597	0.1111	0.7324
Cafeteria (0 dB)	0.1111	0.4786	0.1111	0.4772
Traffic (10 dB)	0.4017	0.7596	0.1746	0.7877
Traffic (0 dB)	0.1564	0.7202	0.1111	0.6665

Table 1 - Macro-F1 scores for UniSpeech-SAT and WavLM under clean and six DEMAND noise conditions. Each condition contains 112 unseen-speaker recordings; higher is better.

On clean speech, WavLM achieved a macro-F1 of 0.8186 compared with 0.7573 for UniSpeech-SAT, an advantage of 0.0613. Both clean-trained systems deteriorated when environmental noise was introduced, although the extent depended strongly on the noise source. Under cafeteria noise at both 10 dB and 0 dB, the macro-F1 of each clean-trained model fell to 0.1111. In kitchen noise at 10 dB, UniSpeech-SAT retained a higher score (0.3686) than WavLM (0.2447), showing that the clean-speech ranking did not persist across all acoustic conditions.

Further examination revealed that WavLM experienced similarly severe degradation under other noise conditions. Under traffic noise at 0 dB, its clean-trained classifier achieved a macro-F1 of only 0.1111 and an accuracy of 28.57%. Kitchen noise at 0 dB produced similarly poor performance, with macro-F1 falling to 0.1278 and accuracy to 29.46%. These findings suggest that WavLM's clean-trained classifier was particularly vulnerable to severe acoustic interference, despite achieving the highest performance on clean speech.

After noise-augmented training, macro-F1 improved in all six noisy conditions for both representations. *Table 2* reports the absolute changes in score. The largest improvement for WavLM was +0.6213 under cafeteria noise at 10 dB, while the largest for UniSpeech-SAT was +0.5638 under traffic noise at 0 dB. Clean-speech performance also improved slightly, by +0.0317 and +0.0419 for WavLM and UniSpeech-SAT respectively. These values are differences in F1 score, not percentage increases.

Audio condition	UniSpeech-SAT Change in macro-F1	WavLM Change in macro-F1
Clean	+0.0419	+0.0317
Kitchen (10 dB)	+0.3829	+0.5586
Kitchen (0 dB)	+0.3908	+0.5777
Cafeteria (10 dB)	+0.5486	+0.6213
Cafeteria (0 dB)	+0.3674	+0.3661
Traffic (10 dB)	+0.3579	+0.6131
Traffic (0 dB)	+0.5638	+0.5554

Table 2 - Absolute change in macro-F1 after noise-augmented training, calculated as augmented minus clean-trained score. Positive values indicate improvement.

The effect of augmentation was not uniform. At 10 dB SNR, augmented WavLM outperformed augmented UniSpeech-SAT in kitchen, cafeteria and traffic noise. At 0 dB SNR, UniSpeech-SAT was stronger in kitchen and traffic noise, while the two models were almost equal in cafeteria noise (0.4786 and 0.4772). For UniSpeech-SAT, the kitchen score was slightly higher at 0 dB than at 10 dB, so degradation was not strictly monotonic with SNR in this fixed test sample.

Figure 1 illustrates the large difference between clean-trained and noise-augmented classifiers for each encoder. The same acoustic conditions and scoring scale are used in both panels, allowing the direction and magnitude of the changes to be compared visually.

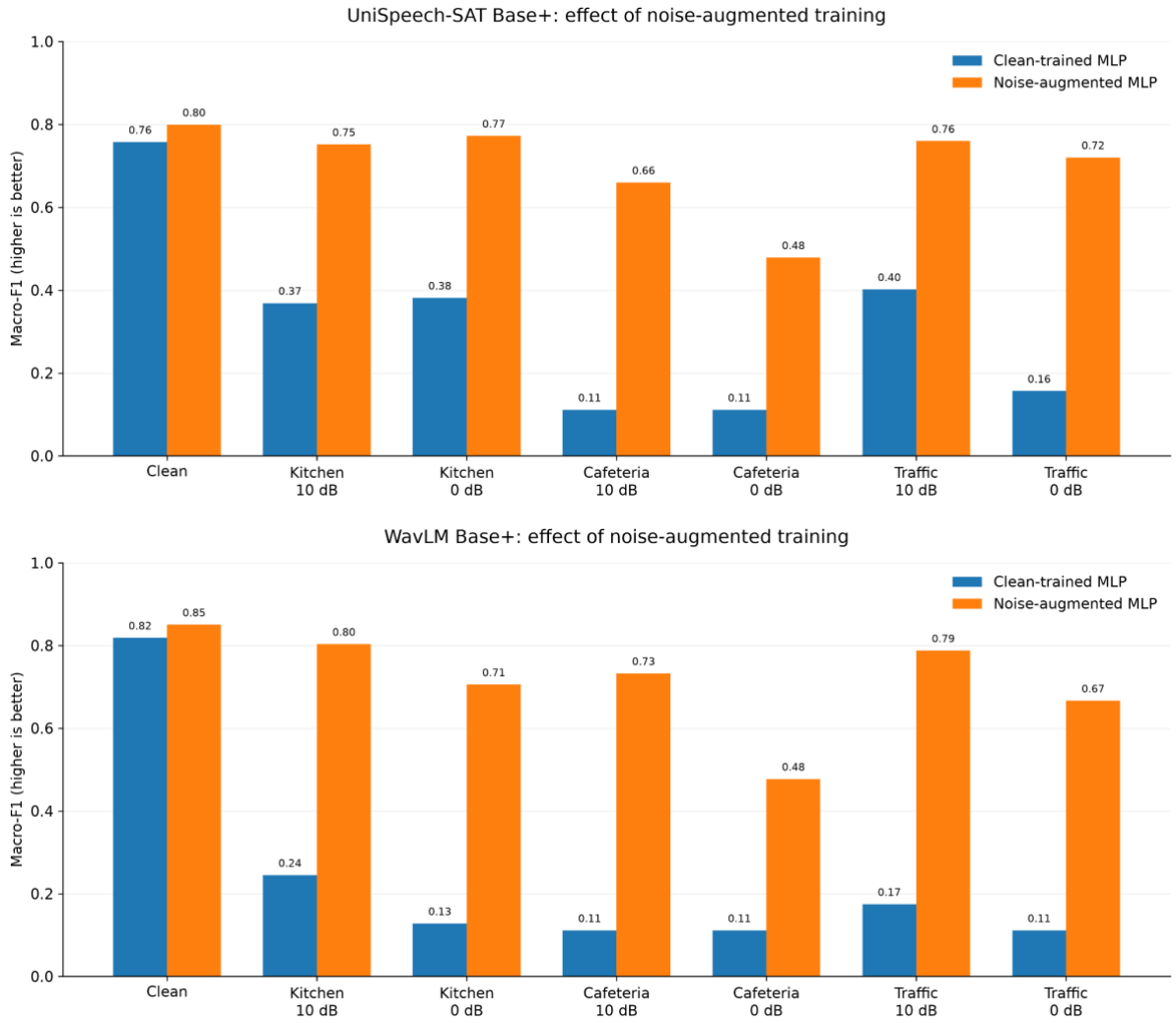


Figure 1 - Macro-F1 before and after noise-augmented MLP training for UniSpeech-SAT Base+ (upper panel) and WavLM Base+ (lower panel), across clean and noisy test conditions.

Classification accuracy provides a complementary measure of performance (*Table 3*). The clean-trained WavLM classifier correctly classified 92 of 112 clean test samples (82.14%), compared with 86 of 112 (76.79%) for UniSpeech-SAT. Under cafeteria noise at 10 dB, both clean-trained models recorded 28.57% accuracy, rising to 74.11% for augmented WavLM and 69.64% for augmented UniSpeech-SAT. The difference between accuracy and macro-F1 in the degraded conditions makes the class-level analysis especially important.

Audio condition	UniSpeech-SAT Clean-trained	UniSpeech-SAT Noise-augmented	WavLM Clean-trained	WavLM Noise-augmented
Clean	0.7679	0.8125	0.8214	0.8482
Kitchen (10 dB)	0.4196	0.7679	0.3482	0.8036
Kitchen (0 dB)	0.4464	0.7768	0.2946	0.7054
Cafeteria (10 dB)	0.2857	0.6964	0.2857	0.7411
Cafeteria (0 dB)	0.2857	0.5089	0.2857	0.5179
Traffic (10 dB)	0.4643	0.7679	0.3125	0.7946
Traffic (0 dB)	0.3036	0.7321	0.2857	0.7054

Table 3 - Classification accuracy across clean and noisy test conditions. Values are proportions of correctly classified recordings (n = 112 per condition).

Cafeteria noise at 10 dB was selected for detailed class-level analysis because both clean-trained classifiers exhibited complete prediction collapse, while noise-augmented training produced substantial improvements. Before augmentation, both classifiers assigned every one of the 112 test recordings to the sad category, correctly identifying only the 32 recordings whose true label was sad. Consequently, accuracy remained 0.2857 despite the absence of meaningful discrimination among the other three emotions. *Table 4* shows that the corresponding F1 scores for neutral, happy and angry were all zero, whereas the sad F1 was 0.4444.

Emotion	UniSpeech-SAT Clean-trained	UniSpeech-SAT Noise-augmented	WavLM Clean-trained	WavLM Noise-augmented
Neutral	0.0000	0.4615	0.0000	0.7097
Happy	0.0000	0.6667	0.0000	0.6102
Sad	0.4444	0.7654	0.4444	0.8286
Angry	0.0000	0.7451	0.0000	0.7812

Table 4 - Per-emotion F1 scores under cafeteria noise at 10 dB SNR for clean-trained and noise-augmented classifiers on the same 112 test recordings.

The confusion matrices in Figure 2 show how this behaviour changed following augmentation. The augmented WavLM classifier correctly identified 11 of 16 neutral, 18 of 32 happy, 29 of 32 sad and 25 of 32 angry recordings (83 correct in total). UniSpeech-SAT correctly identified 6, 22, 31 and 19 recordings respectively (78 in total). Predictions were again distributed across all four categories, although the remaining errors differed: UniSpeech-SAT frequently labelled neutral as sad and angry as happy, while WavLM confused several happy and angry examples.

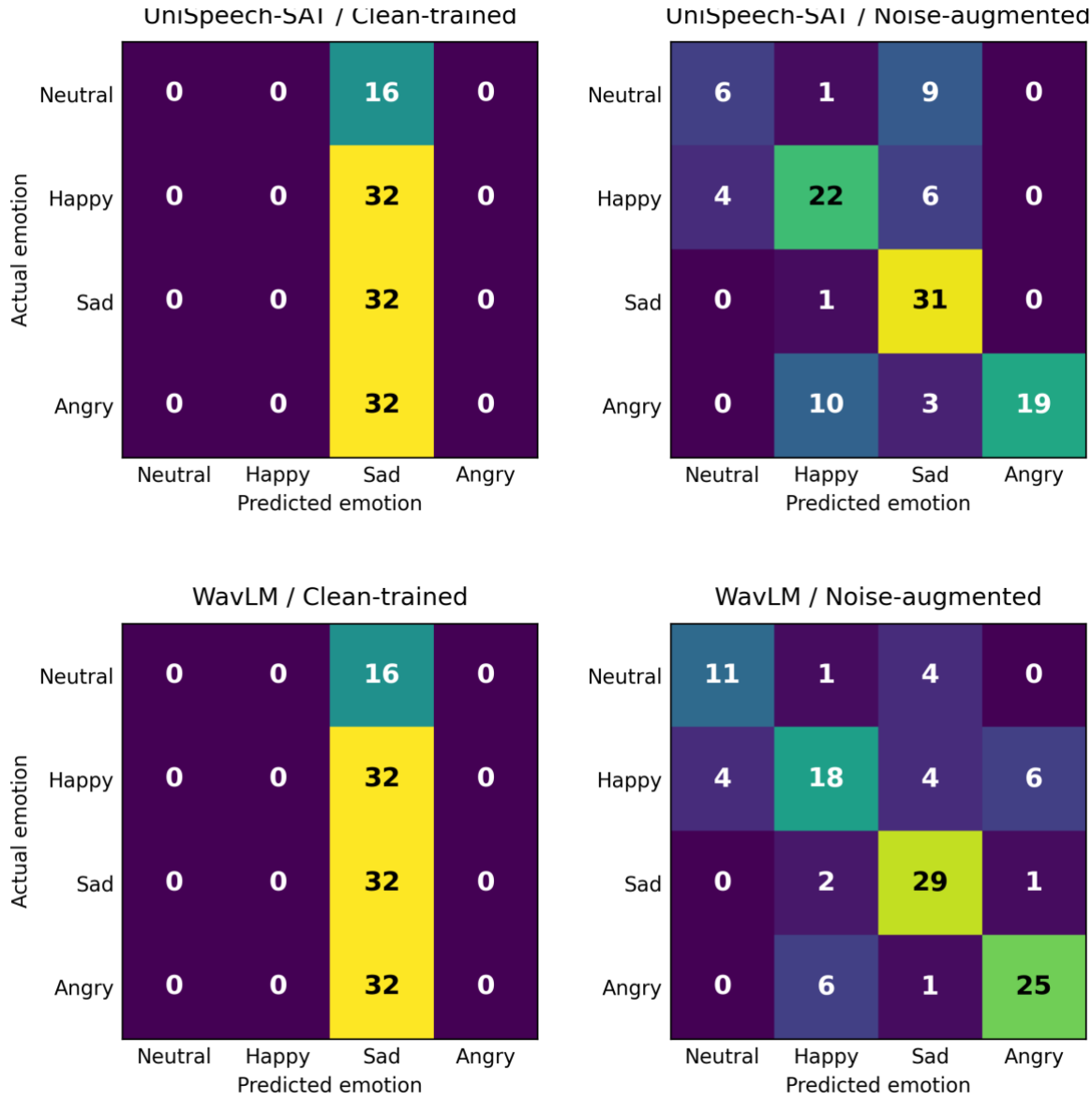


Figure 2 - Confusion matrices for cafeteria noise at 10 dB SNR: clean-trained and augmented UniSpeech-SAT (top row), followed by clean-trained and augmented WavLM (bottom row). Rows indicate true labels; columns indicate predictions. Cell values are counts.

As a secondary computational observation, extracting embeddings for all 672 clean recordings took approximately 90.0 s for UniSpeech-SAT and 110.5 s for WavLM in the recorded local runs, equivalent to about 0.134 s and 0.164 s per recording respectively. These are implementation-specific processing times, not a controlled hardware benchmark. Collectively, the results demonstrate substantial sensitivity of the original clean-trained classifiers to acoustic interference and consistent improvements after downstream noise augmentation. The same three DEMAND recordings were used to prepare augmented training examples and noisy tests, so these gains establish improvement under the tested noise sources rather than generalisation to entirely unseen environmental recordings.

In relation to RQ1, WavLM achieved stronger clean-speech performance, but under environmental noise, the clean-trained UniSpeech-SAT classifier achieved equal or higher macro-F1 scores in all six conditions. This indicates that WavLM's advantage on clean speech did not translate into greater robustness under the tested noise conditions. Following noise-augmented training, WavLM performed better across all three environments at 10 dB, whereas UniSpeech-SAT performed better at 0 dB, although the cafeteria results were nearly identical. These differences should be interpreted cautiously given the limited test sample. Regarding RQ2, noise-augmented classifier training improved macro-F1 across all six noisy conditions for both representations without modifying the pretrained encoders, demonstrating improved robustness to the tested noise sources rather than necessarily to unseen acoustic environments.

5) Discussion

5.1) Summary of findings. This study asked how frozen UniSpeech-SAT Base+ and WavLM Base+ representations compare in clean and noisy speech emotion recognition on unseen speakers (RQ1), and whether noise-augmented classifier training reduces noise-related degradation without modifying the encoders (RQ2). For RQ1, WavLM was stronger on clean speech (macro-F1 0.8186 vs 0.7573, Table 1), but this advantage did not carry over to noise. The clean-trained UniSpeech-SAT classifier scored equal to or higher than WavLM in all six noisy conditions (mean macro-F1 0.255 vs 0.147), and the ranking after augmentation depended on SNR and noise type. For RQ2, augmentation improved macro-F1 in all six noisy conditions for both models (Table 2), raising the six-condition mean to 0.690 for UniSpeech-SAT and 0.695 for WavLM. RQ2 is therefore answered more firmly than RQ1: its effect has a consistent direction in every condition, whereas the RQ1 result is a pattern of mixed rankings on a small test set.

5.2) Clean-speech performance did not predict noisy performance. WavLM's clean-speech lead of 0.061 is in the direction expected from [5], where both models are among the strongest SSL representations for SER. However, with 112 recordings from four speakers, a 95% interval on an accuracy near 0.8 is roughly ±7–8 points, and wider still because recordings from one speaker are not independent. The lead should therefore be treated as suggestive, not established. The noisy-condition differences are larger than this uncertainty, for example kitchen noise at 0 dB (0.3813 vs 0.1278) and traffic noise at 10 dB (0.4017 vs 0.1746). Denoising during pretraining [2] therefore did not translate into robustness in this setup, which used final-layer mean-pooled embeddings and a classifier trained only on clean speech. One untested explanation is that WavLM's denoising objective is aimed at recovering speech content and speaker information, which need not keep emotion-relevant cues stable, while UniSpeech-SAT's utterance mixing [1] exposes it to speech-on-speech interference. Testing this would need layer-wise or embedding-shift analysis. Because the models also differ in training data and architecture, the differences cannot be attributed to their pretraining objectives.

5.3) Failure mode of clean-trained classifiers. Under cafeteria noise at 10 dB, both clean-trained classifiers assigned all 112 recordings to "sad". Accuracy stayed at 0.2857 (the sad

class is 32 of 112) while macro-F1 fell to 0.1111, with zero F1 for the other three emotions (Table 4, Figure 2). This was not a one-off. Clean-trained WavLM also collapsed at traffic 0 dB (accuracy 0.2857, macro-F1 0.1111) and nearly so at kitchen 0 dB (0.2946, 0.1278), and clean-trained UniSpeech-SAT collapsed in both cafeteria conditions. Because the same frozen encoders supported macro-F1 of 0.73 and 0.66 once the classifiers were trained on noisy embeddings, emotion-relevant information appears to have remained at least partly available in the representations. This suggests that a mismatch between the clean-trained decision boundary and the noisy embedding distribution may have contributed to the failure, rather than a complete loss of emotion-relevant information within the representations. A plausible but untested mechanism is that noise displaces the mean-pooled embeddings so that, after standardisation with clean-training statistics, inputs fall in a region the classifier maps to a single class. Why that class is "sad" is not explained by these results. The inference also has a limit: the augmented classifiers saw the same noise recordings used in testing, so some of the recovery could reflect noise-specific cues.

5.4) Effect of noise-augmented training. Gains ranged from +0.03 (clean) to +0.62 (WavLM, cafeteria 10 dB: 0.1111 to 0.7324). The mean gain across the six noisy conditions was +0.44 for UniSpeech-SAT and +0.55 for WavLM. These changes are far larger than any difference between the models after augmentation (mean 0.695 vs 0.690), so in this setup the downstream training condition influenced robustness more than the choice of encoder. This is consistent with augmentation helping SER systems built on conventional features [7], and with the broader principle in [15] that exposing a model to target-condition data reduces mismatch (there during ASR pretraining, here at the classifier stage). It differs from [12], which adds environment information and adapts the SER model, whereas no environment information was used here and only the tested noise sources were evaluated. Cafeteria noise at 0 dB remained hardest (0.4786 vs 0.4772). Babble-like interference overlapping speech in spectrum and modulation may be harder to separate, although this was not tested. After augmentation, the remaining confusions were between neutral and sad and between happy and angry (Figure 2), which is consistent with the arousal-related confusions commonly reported in SER. Augmentation also improved clean-speech performance (+0.032 and +0.042), which suggests part of the gain is a general data or regularisation effect, not specific to noise.

5.5) Relation to previous work. Direct numerical comparison with published RAVDESS or noisy-SER results would be misleading, because splits, class subsets and metrics differ, and [12] predicts continuous attributes. Qualitatively, the findings agree that lightweight classifiers on SSL embeddings work well on clean speech [5], [6], and that augmentation helps [7]. The new observation is that a representation ranking from clean speech [5] did not carry over to controlled noise for these two Base+ models, and that clean-trained classifiers can collapse completely while overall accuracy hides it.

5.6) Limitations. The design used one actor-independent split, with four test speakers and four validation speakers. Each configuration was trained once (seed 42) and no confidence intervals or significance tests were computed, so differences of a few points should be read cautiously. RAVDESS is acted speech, and noise was mixed synthetically without reverberation or microphone effects. The same three DEMAND recordings were used for augmentation and testing, so gains reflect the tested noise sources, not unseen environments.

Augmentation tripled the training set and changed validation-based model selection, so noise exposure cannot be separated from data volume. Finally, only final-layer embeddings with mean pooling were used, although intermediate layers may carry more emotion information [11].

5.7) Future work. In order of value: (i) leave-one-environment-out training and testing on held-out noise; (ii) several speaker splits and seeds, reported with bootstrap confidence intervals or paired tests; (iii) a size-matched control for augmentation; (iv) layer-wise representations and attention-based pooling [11]; and (v) an embedding-shift analysis to test the proposed collapse mechanism.

Conclusion:

This study compared frozen UniSpeech-SAT Base+ and WavLM Base+ representations for four-class speech emotion recognition on unseen RAVDESS speakers, in clean and DEMAND-noise conditions. In answer to RQ1, WavLM was stronger on clean speech (macro-F1 0.8186 vs 0.7573), but clean-trained UniSpeech-SAT scored equal to or higher in all six noisy conditions. Several clean-trained classifiers collapsed onto a single class (macro-F1 as low as 0.1111), a failure that accuracy alone concealed. In answer to RQ2, training the classifiers on clean and noise-augmented embeddings improved macro-F1 in all six noisy conditions (by up to +0.62 for WavLM and +0.56 for UniSpeech-SAT) without modifying the encoders. The ranking of the two models after augmentation still depended on SNR and noise type. These conclusions are limited by a single speaker split with four test speakers, one training seed, and noise recordings shared between augmentation and testing, so they describe the tested conditions, not generalisation to unseen environments. Held-out noise, multiple speaker splits with confidence intervals, and a size-matched control are the most informative next steps.

GitHub Link: https://github.com/SajjadIbrahim123/elec5305-project-540861035

Note: The RAVDESS and DEMAND audio datasets have not been uploaded to the GitHub repository due to their large storage requirements. Instead, official download links and instructions for organising the datasets are provided in the repository's README file. The repository contains the Python implementation, experimental results, tables and figures. The original audio recordings extracted embeddings and trained classifier checkpoints are excluded and can be generated by following the project workflow.

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Video: https://youtu.be/3LLGBXtNwOM (unlisted - only available for link)

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